Journal of Pharmaceutical Sciences 109 (2020) 3300-3307

Contents lists available at ScienceDirect

Journal of Pharmaceutical Sciences

journal homepage: www.jpharmsci.org

Pharmaceutical Biotechnology

Rapid Polysorbate 80 Degradation by Liver Carboxylesterase in a Monoclonal Antibody Formulated Drug Substance at Early Stage Development

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ARTICLE INFO

Article history:
Received 15 May 2020
Revised 16 July 2020
Accepted 20 July 2020
Available online 25 July 2020

Keywords:
Antibody
Antibody drugs
Hydrolysis
Liquid chromatography-mass spectrometry (LC-MS)
Proteomic

ABSTRACT

Polysorbates (PS) are surfactants commonly added in a therapeutic protein drug product to protect proteins from denaturation and aggregation during storage, transportation, and delivery. Significant degradation of PS in drug products could lead to particulate formation with shortened drug shelf life, and one of the major root causes of PS degradation is the host cell protein (HCP) derived lipase/esterase, which belong to the serine hydrolase family. Typically, PS degradation can only be observed in drug products after a long time of storage if very low levels of host cell protein impurity with PS degradation activities are present. In this study, PS80 degradation was observed in a monoclonal antibody (mAb) within 18 h at 5 °C with a low level of HCP presented (<20 ppm) based on ELISA quantitation. This observation suggested that a trace amount of unknown host cell protein(s) with strong enzymatic activity on polysorbate degradation was present in this drug substance. The activity-based protein profiling (ABPP) method with the ActivX FP serine hydrolase probe was employed to identify host cell proteins that can hydrolyze PS. Two hydrolases, liver carboxylesterase B-1-like protein (CES-B1L, A0A061I7X9) and liver carboxylesterase 1-like protein (CES-1L, A0A061IFE2) were identified with high confidence using the ABPP approach for the first time in a mAb drug substance during early stage development. PS80 became stable in the drug substance sample after the two hydrolases were depleted using the immobilized ActivX FP probe, confirming these two hydrolases were responsible for the rapid PS80 degradation. In addition, the PS80 degradation pattern was found to be equivalent to that generated by their human analog, human liver carboxylesterase-1 (hCES-1) and rabbit liver esterase (rLES). Overall, these results suggest that CES-B1L and CES-1L are the primary cause of PS80 degradation in this mAb drug.

Introduction

PS20 and PS80 are the most commonly used surfactants in biopharmaceutical protein formulation that can improve protein stability and protect drug products from aggregation and denaturation during delivery and transportation.[2,3] However, polysorbate is liable to oxidative degradation[4,5] as well as enzymatic hydrolysis,[6,7] and the degraded products can drive undesired particulate formation in the formulated drug substances. For example, porcine liver esterase was reported to be able to specifically hydrolyze PS80.[8] Lipoprotein Lipase (LPL) was found to have enzymatic activity on both PS20 and PS80.[9] Group XV lysosomal phospholipase A2 isomer X1 (LPLA2) can degrade both PS20 and PS80 at less than 1 ppm.[7,10] The enzyme family carboxylester hydrolases (EC 3.1.1) in different organisms, including the bacteria *Pseudomonas cepacia*, the fungi *Thermomyces lanuginosus* and *Candida antartica* have also been reported to be able to degrade PS20 and PS80.[11]

Polysorbate degradation caused by host cell proteins in drug products can be observed after storage at 5 °C for months or years[5,9] as the impurities are usually below 100 ppm after multiple steps of purification. Under thermal stress condition, the degradation can be accelerated to days. In one mAb drug substance at its early stage development, we observed an unusually rapid PS80 degradation that occurred within several hours at 5 °C. The total level of HCPs was determined to be less than <20 ppm by ELISA immunoassay. Immunoprecipitation enrichment using commercial anti-HCP antibodies followed by nano LC-MS (IP-MS) was performed for this drug substance with no lipase/esterase detected, indicating certain highly active host cell proteins that can degrade PS80 was present below the detection limit or off the coverage of IP-MS method (1–10 ppm) applied to the investigation.[12] Most of the host cell

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https://doi.org/10.1016/j.xphs.2020.07.018
0022-3549/

proteins that degrade PS belong to the serine hydrolases family, i.e. pancreatic lipase (PNLIP), carboxyl ester lipase (CEL), lipoprotein lipase (LPL), phospholipase A2 (PLA2), lysosomal PLA2 (LPLA2) and phospholipase B (PLB1).[13] Members of the serine hydrolases family share a serine centric charge triad in their catalytic site and this common feature can be exploited with ABPP to selectively label active serine hydrolases.[1,14,15] This family of hydrolases can be captured and enriched by ABPP probe when the probe is tagged with biotin for selective enrichment.[1] Unlike general multidimensional protein identification technology (MudPIT), ABPP targets the active enzyme directly, improving the identification of these specific enzymes when they are present at a very low abundance. This feature of ABPP provides a valuable tool for accessing active hydrolases since many of the enzymes may not be active in either their zymogen form or with endogenous inhibitors.

In this study, we found that PS80 degradation was completely inhibited by the commercially available ActivX FP serine hydrolase probe. The ActivX FP serine hydrolase was proved to be highly specific to serine hydrolases.[14–17] We immobilized the probe by binding the biotin-modified probe to the streptavidin derived magnetic beads and performed the depletion experiments. The serine hydrolase depleted mAb drug substance lost its activity to degrade PS80. The hydrolases that were captured and enriched by the ABPP approach were also subjected to nano LC-MS/MS analysis. Two closely related serine hydrolases, CES-B1L and CES-1L, were identified. To confirm the function of the enzymes, recombinant hCES-1, and rabbit liver esterase (rLES), which shares similar sequences as CES-B1L and CES-1L, was incubated with PS80 to compare the degradation pattern.

Materials and Methods

Materials

Dynabeads MyOne Streptavidin T1 was purchased from Invitrogen of Thermo Fisher Scientific (Waltham, MA). ActivX Desthiobiotin-FP Serine Hydrolase Probe, Formic acid, acetonitrile, Diothiothreitol (DTT) and 1-step ultra TMD-blotting solution were purchased from Thermo Fisher Scientific (Waltham, MA). Acetic acid, 10× Tris buffered saline (TBS), Iodoacetamide (IAM), bovine serum albumin (BSA) and urea were purchased from Sigma-Aldrich (St. Louis, MO). HEPES buffered saline with EDTA and 0.005% v/v Surfactant P-20 (HBS-EP) were purchased from GE (Boston, MA). Monoclonal antibody drug substance was made at Regeneron Pharmaceutical Inc. Polysorbate 80 was purchased from Croda (East Yorkshire, UK). CHO HCP ELISA Kit (F550) was purchased from Cygnus Technologies. Rabbit rLE was purchased from Sigma Aldrich (St. Louis, MO). Human CES-1 was purchased from Abcam (Cambridge UK). Sequencing Grade Modified Trypsin was purchased from Promega (Madison, WI). CHO recombinant transthyretin and CHO recombinant peroxiredoxin-1 were produced in house. Annexin was purchased from GenScript. Oasis Max column (2.1 × 20 mm, 30 μm) and Acquity UPLC BEH C4 column (2.1 × 50 mm, 1.7 μm) were purchase from Waters (Milford, MA). Acclaim PepMap 100C18 analytical column (0.075 × 250 mm, 3 μm) and Acclaim PepMap 100C18 trap column (0.075 × 20 mm, 3 μm) were purchased from Thermo Fisher Scientific (Waltham, MA). DPBS (10×) was purchased from Gibco (Thermo Fisher Scientific, Waltham, MA) and Tween 20 was purchased from J.T. Baker (Phillipsburg, NJ). Q-Exactive Plus with electrospray ionization (ESI) source was purchased from Thermo Fisher Scientific (Waltham, MA). mAb-1 is an IgG4 antibody (50 mg/mL) formulated in histidine buffer with sucrose and PS80 as excipients.

Two-Dimensional Liquid Chromatography- Charged Aerosol Detection (CAD)/Mass Spectrometry (MS) Method to Analyze Polysorbate Degradation

The degradation of PS80 in CHO cell-free media or formulated antibody were analyzed by two-dimensional HPLC-CAD/MS method as previously described by Li et al.[18] Polysorbates were first separated from formulated mAb using Oasis MAX column (2.1 × 20 mm, 30 μm) pre-equilibrated with 99% solvent A (0.1% formic acid in water) and 1% solvent B (0.1% formic acid in acetonitrile). Post sample injection, the equilibration gradient was held for 1 min, followed by a linear increase of Solvent B to 15% in 4 min to separate polysorbate from mAb. The eluted polysorbates were then diverted to Acquity BEH C4 column (2.1 × 50 mm, 1.7 μm) using a switch valve for reversed phase chromatography-based separation. At the start of the separation, solvent B was quickly increased to 20% in 1.5 min, then gradually increased to 99% at 45 min and held for 5 min, followed by an equilibration step of 1%B for 5 min. The flow rate was kept at 0.1 mL/min and column temperature at 40 °C.

The 2D-LC system was set up with Thermo UltiMate 3000 and coupled with Corona Ultra CAD detector, operating at nitrogen pressure of 75 psi for quantitation. Chromeleon 7 was used for system control and data analysis. Q-Exactive Plus with ESI source was coupled with the 2D-LC system for characterization only. The instrument was operated in a positive mode with capillary voltage at 3.8 kV, capillary temperature at 350 °C, sheath flow rate at 40, and aux flow rate at 10. Full scan spectra were collected over the *m/z* range of 150–2000. Thermo Xcalibur software was used to collect and analyze MS data.

Peak area of each ester was obtained from the CAD chromatogram and added up to account for intact PS80. The remaining percentage of PS80 after degradation was calculated by comparing the sum of the peak area of monoester eluting between 25 min and 30 min at each time point to the sum of peak areas at time zero. Relative percentage of different order ester or total esters can be calculated similarly.

PS80 Degradation Assay with Human CES-1, Rabbit LES and Formulated Antibody

The effect of human CES-1 and rabbit LES on PS80 was examined by mixing 2 μL 0.1 mg/mL human CES-1 or 0.02 mg/mL rabbit LES with 2 μL 1% PS80 in 16 μL 10 mM histidine buffer, pH 6.0, followed with incubation at 4 °C for 1.5, 8 and 18 h, respectively. One aliquot (3 μL) of each solution was diluted 25 times by adding 72 μL 10 mM histidine, pH 6.0, before the LC-CAD analysis.

The hydrolysis of PS80 in formulated mAb was examined by mixing 18 μL 50 mg/mL mAb (buffer exchange to 10 mM histidine, pH 6.0) with 2 μL 1% PS80 then incubated at 5 °C for 18, 24 and 36 h. One aliquot (3 μL) of each solution was diluted 25 times with 10 mM histidine, pH 6.0 before the LC-CAD analysis.

Inhibition of Serine Hydrolase from CHO-Derived Antibodies

ActivX Desthiobiotin-FP serine hydrolase probe were diluted in DMSO to 0.1 mM as stock solution. Aliquoted 1.25 μL, 5 μL and 20 μL probe stock solution each was mixed with 5 mg mAb in 1 × PBS in a final volume of 1 mL, followed by gentle rotating at room temperature for 1 h. Each mixture was then buffer exchanged into 10 mM histidine, pH 6.0 to remove the free probes, and the mAb concentration was adjusted to 50 mg/mL. Each buffer exchanged sample was incubated with 0.1% PS80 at 5 °C followed by LC-CAD PS80 degradation assay.

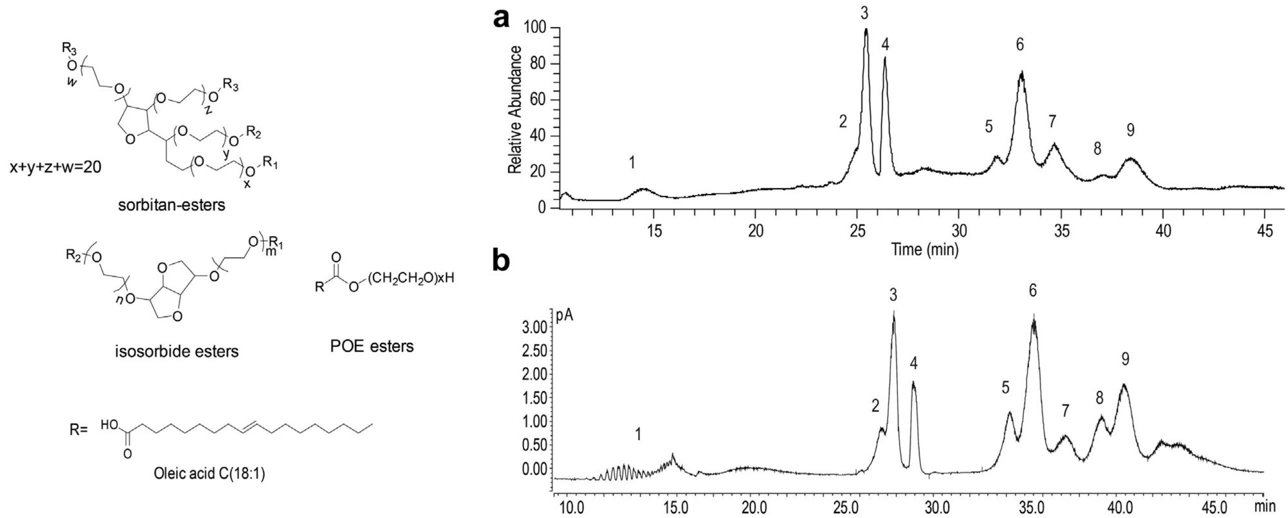

Fig. 1. Left panel: chemical structure of major species in polysorbates. Polysorbates are mainly composed of fatty acid esters sharing a common sorbitan POE, isosorbide POE or POE head group, with oleic acid as the main fatty acid for PS80. Right panel: A): Total ion current (TIC) chromatogram of PS80 in mAb formulation by online 2D-LC/MS analysis. The identity of the labeled peaks are: (1) POE- POE isosorbide- POE sorbitan, (2) POE sorbitan monolinoleate, (3) POE sorbitan monooleate, (4) POE isosorbide monooleate and POE monooleate, (5) POE sorbitan linoleate/oleate diester, (6) POE sorbitan di-oleate, (7) POE isosorbide di-oleate and POE di-oleate, (8) Probably POE isosorbide/POE linoleate/oleate diester as mass spectra are too complicated to interpret, (9) POE sorbitan mixed trioleate and tetraoleate. Right panel (B): CAD chromatogram showing the separation and detection of PS80 in mAb formulation by online 2D-LC/CAD analysis.

Depletion of Serine Hydrolase from CHO-Derived Antibodies

Serine hydrolase depletion experiment was performed by using immobilized ActivX Desthiobiotin-FP serine hydrolase probe. To immobilize the probe, 35 μL ActivX Desthiobiotin-FP serine hydrolase probe (0.1 mM stock solution in DMSO) was first coupled with 2 mg Streptavidin Dynabeads to a final volume of 1 mL in 1 × PBS by gentle rotating at room temperature for 2 h. Process control sample was prepared by mixing 35 μL DMSO with 2 mg Streptavidin Dynabeads to a final volume of 1 mL in 1 × PBS and gentle rotating at room temperature for 2 h. The beads were washed by 1 mL 1 × PBS 3 times and then resuspended into 800 μL 1 × PBS. 5 mg mAb sample was then added into the FP probe-coupled Streptavidin Dynabeads and incubated at room temperature with gentle rotation for 1 h. The supernatant was buffer exchanged into 10 mM histidine, pH 6.0, and the mAb concentration adjusted to 50 mg/mL. The buffer-exchanged supernatant samples were then incubated with 0.1% PS80 at 5 °C followed by LC-CAD PS80 degradation assay.

Detection of Host Cell Proteins (HCPs) in CHO-Derived Antibodies with ABPP

ActivX Desthiobiotin-FP serine hydrolase probe were diluted in DMSO to 0.1 mM as stock solution. Aliquoted 20 μL probe stock solution was first mixed with 5 mg mAb in 1 × PBS to a final volume of 1 mL, followed by gentle rotating at room temperature for 1 h. Free probes were removed by filtration and protein was recovered by 5 M urea in PBS. 2 mg Streptavidin Dynabeads was added to the solution and incubated by gentle rotating at room temperature for 2 h. After removing the supernatant, Dynabeads were collected by magnet, and washed by 0.5 mL 5 M urea in PBS and then resuspend into 100 μL 5 M urea/50 mM tris solution with 5 mM TCEP. The proteins were denatured and reduced at 55 °C for 30 min and then incubated with 10 mM iodoacetamide for 30 min in dark. Alkylated proteins were diluted 5 times and digested with 1 μg trypsin at 37 °C for overnight. Dynabeads were removed by magnet and the supernatant with peptide mixture was acidified by 5 μL of 10% FA, desalted using GL-Tip™ SDB desalting tip (GL science, Japan) and resuspended into 40 μL 0.1% FA. 15 μL were transferred to Eppendorf tubes for Nano LC-MS/MS analysis and the rest were stored at −80 °C. Negative control was performed by heating mAb sample at 80 °C for 5 min first to denature all proteins to prevent host cell proteins from binding to the ActivX Desthiobiotin-FP serine hydrolase probe.

LC-MS/MS Analysis

The peptide mixture was dissolved in 40 μL of 0.1% formic acid (FA) and 10 μL was first loaded onto a 20 cm × 0.075 mm Acclaim PepMap 100C18 trap column (Thermo Fisher Scientific) for desalting and later separated on a 250 mm × 0.075 mm Acclaim PepMap

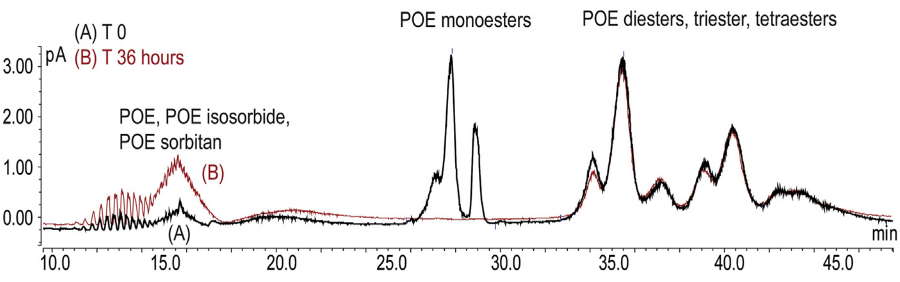

Fig. 2. Chromatogram of 0.1% PS80 in 50 mg/mL mAb-1 incubated at 5 °C in 10 mM Histidine, pH 6 for 0 h and 36 h.

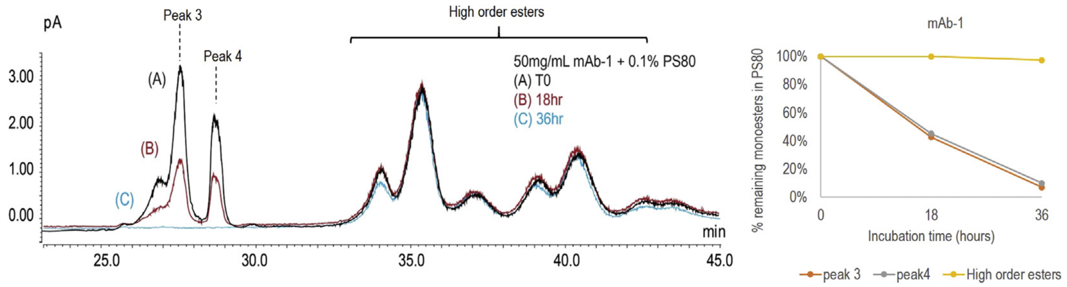

Fig. 3. Left Panel: chromatogram of 0.1% PS80 in 50 mg/mL mAb-1 incubated at 5 °C in 10 mM Histidine, pH 6 for 0, 18 and 36 h. Right Panel: Percentage of 0.1% remaining after 18 and 36 h of incubation with 50 mg/mL mAb-1. Percentage of the initial area remaining for the representative peak 3 (POE sorbitan monooleate), peak 4 (POE isosorbide monooleate and POE monooleate) and sum of higher order esters (di-, tri-, tetra-esters) after hydrolysis is shown with orange, gray and yellow, respectively.

100C18 analytical column in an UltiMate 3000 nanoLC (Thermo Fisher Scientific). The mobile phase A was made of 0.1% FA in ultra-pure water and mobile phase B was made of 0.1% FA in 80% ACN. The peptides were separated with a 150 min linear gradient of 2%–32% of buffer B at flow rate of 300 nL/min. The UltiMate 3000 nanoLC was coupled with a Q-Exactive HFX mass spectrometer (Thermo Fisher Scientific). The mass spectrometer was operated in the data-dependent mode in which the 10 most intense ions were subjected to higher-energy collisional dissociation (HCD) fragmentation with the normalized collision energy (NCE) 27%, AGC 3e6, max injection time 60 ms) for each full MS scan (from *m/z* 375–1500 with resolution of 120,000) and AGC 1e5, max injection time 60 ms for MS/MS events (from *m/z* 200–2000 with resolution of 30,000).

mAb-1 Direct Digestion

100 μg of mAb-1 was dried with speed vacuum, then reconstituted with 20 μL 8 M urea containing 10 mM DTT. The protein was denatured and reduced at 55 °C for 30 min, and then incubated with 6 μL of 50 mg/mL iodoacetamide for 30 min in dark. Alkylated protein was digested with 100 μL 0.1 μg/μL trypsin at 37 °C for overnight. The peptide mixture was acidified by 5 μL of 10% TFA. The sample was diluted to 0.4 μg/μL and 2 μL was injected onto the column for LC-MS/MS analysis.

PRM Analysis of CES-B1L and CES-1L in mAb-1

Direct digestion samples (0.8 μg) were loaded onto a 20 cm × 0.075 mm Acclaim PepMap 100C18 trap column (Thermo Fisher Scientific) for desalting and later separated on a 250 mm × 0.075 mm Acclaim PepMap 100C18 analytical column in an UltiMate 3000 nanoLC (Thermo Fisher Scientific). The column was preequilibrated with 98% mobile phase A (made of 0.1% formic acid in water) and 2% mobile phase B (made of 0.1% formic acid in 80% ACN) at a flow rate of 300 nL/min. Post sample injection a linear gradient from 2% to 37% mobile phase B was applied over 100 min to separate the peptides. Mass spectrometry data were acquired by parallel reaction monitoring (PRM) targeting 3 peptides LNVQGDTK [*m/z* 437.7351^{2+}], AISESGVILVPGLFTK [*m/z* 815.9744^{2+}] and ENHAFVPTVLDGVLLPK [*m/z* 925.0145^{2+}] from CES-1L, 3 peptides APEEILAEK [*m/z* 500.2715^{2+}], DGASEEETNLSK [*m/z* 640.2861^{2+}] and IRDGVLDILGDLTFGIPSVIVSR [*m/z* 819.1355^{3+}] from CES-B1L, and 3 peptides GPSVFPLAPCSR [644.3293^{2+}], LLIYDASNRPTGIPAR [586.3283^{3+}] and STSESTAALGCLVK [712.3585^{2+}] from mAb-1. In all experiments, a full mass spectrum at 120,000 resolution relative to *m/z* 200 (AGC target 1e6, 60 ms maximum injection time, *m/z* 350–2000) was followed by time scheduled PRM scans at 30,000 resolution (AGC target 1e5, 100 ms maximum injection time). Higher-energy collisional dissociation (HCD) was used with 27% normalized collision energy and an isolation window of 2 *m/z* for MS/MS analysis.

Results

Polysorbate in mAb Formulation Detected by 2D-LC-CAD/MS

Polysorbate in formulated mAbs was separated, identified, and quantitated by 2D-LC-CAD/MS.[18,19] The first dimensional LC by Oasis Max column was designed to remove mAb, and the second dimensional reversed phase chromatography was implemented to separate the remaining POE and POE esters based on their fatty acid content and type. The PS80 species eluted in the order of POE, POE isosorbide, POE sorbitan, monoesters, diesters, triesters and tetraesters (Fig. 1, right panel). The structure of each ester was elucidated by mass spectrometry based on the chemical formula of the polymer and dioxolanylium ion generated by in-source

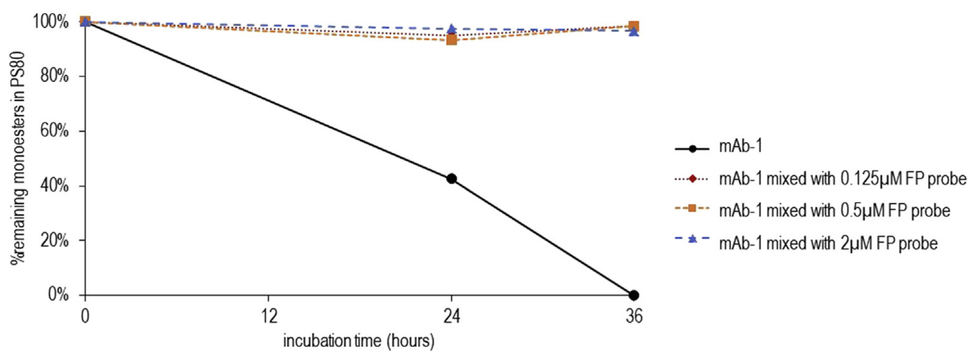

Fig. 4. Inhibition experiments, the original mAb-1, mAb-1 mixed with 0.125 μM, 0.5 μM and 2 μM FP probe are indicated by filled circle with black solid line, filled diamond with red dotted line, filled square with orange dashed line and filled triangle with blue dotted line. The percentage of PS80 remaining is plotted against incubation time.

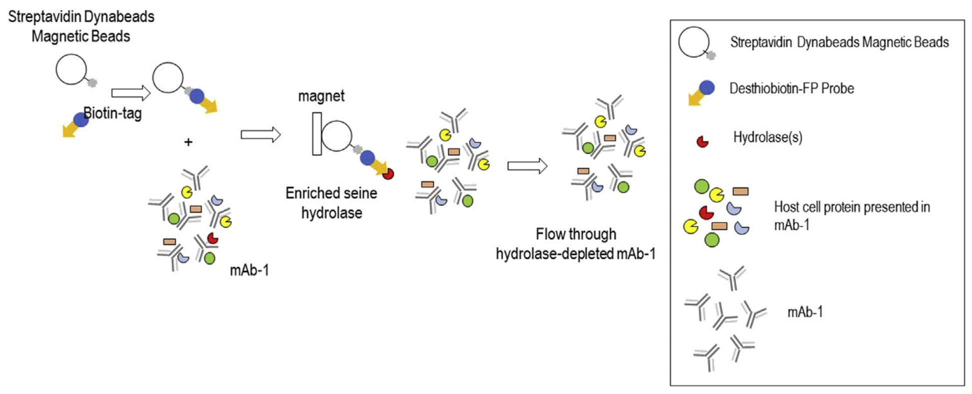

Fig. 5. Schematic diagram of the lipase(s) depletion experiment. Streptavidin dynabeads magnetic beads were coupled with desthiobiotin-FP probe and used for lipase(s) depletion. The original mAb-1 and flow through mAb-1 as well as processed control mAb-1 were incubated with 0.1% PS80 at 5 °C for 36 h and subjected to PS degradation measurement.

fragmentation. Fig. 1 right panel A is the representative total ion current (TIC) chromatogram of PS80 with major peaks labeled, in the eluting order of POE- POE isosorbide- POE sorbitan, POE sorbitan monolinoleate, POE sorbitan monooleate, POE isosorbide monooleate and POE monooleate, POE sorbitan linoleate/oleate diester, POE sorbitan di-oleate, POE isosorbide di-oleate and POE di-oleate, Probably POE isosorbide/POE linoleate/oleate diester and POE sorbitan mixed trioleate and tetraoleate. It should be noted that peak 8 in Fig. 1 was labeled as possible POE isosorbide/POE linoleate/oleate diester mixer as the mass spectra were too complicated to interpret. Quantitation of polysorbates was determined by charged aerosol detection (CAD) chromatography analysis (Fig. 1, right panel B).

Rapid PS80 Degradation in mAb-1 Formulation

Rapid PS80 degradation was observed in mAb-1 during storage at 5 °C for 36 h. Significant decreases occurred in peaks eluting between 25 and 30 min, representing POE monoesters, i.e., POE sorbitan monolinoleate, POE sorbitan monooleate, POE isosorbide monooleate and POE monooleate, while POEs eluting between 10 and 18 min showed significant increases (Fig. 2). There were no changes on POE di-, tri- and tetra-esters eluting between 32 and 45 min. As suggested by a recent elegant study by McShan et al., PS degradation rate is highly dependent on the chemical structures and the order of esters (mono-, di- or trimester). The pattern of PS degradation is unique to hydrolytic enzyme,[11] for example, the degradation rate for POE sorbitan monoester, isosorbide monoester, diesters and triesters in both PS20 and PS80 were all different when treated with different enzymes. Therefore, they suggest that enzyme-mediated PS degradation profile can be served as the fingerprint to confirm the presence of a contaminating enzyme either introduced during expression or carried through the purification process. As shown Fig. 3, besides there was no difference in degradation rate for di-, tri- and tetra-esters, the degradation rates of POE monoesters (peak 3 and 4 in Fig. 1 representing POE sorbitan monooleate and POE isosorbide monooleate, respectively) were identical. The unique degradation pattern observed in our experiments suggests that it can be used as fingerprint to validate the host cell proteins that were responsible for PS80 degradation. These host cell proteins can only degrade the monoester part of PS80 while leaving higher order esters untouched.

Inhibition of Serine Hydrolase by Desthiobiotin- Fluorophosphonate (FP) Probe Results in a Vanished PS80 Degradation

Because most of the host cell proteins that have been reported to degrade polysorbates belong to the family of serine hydrolases,[5,6,8,9] we carried out inhibition experiments using the FP probe that specifically bind and inhibit serine hydrolase activity. This experiment enables the identification and distinction of enzymatically active hydrolyses from other inactive hydrolyses either in their zymogen form or with endogenous inhibitors. The rationale of this experiment is that if there is any active serine hydrolase, adding its inhibitor would stop the enzyme from functioning, in our case, degrading PS80. Desthiobiotin-FP probe is one of the commercially available serine hydrolase probes that contain the reactive fluorophosphonate group which forms covalent bond with Ser at the catalytic center of the serine type hydrolase and blocks its enzymatic activity. The inhibition experiment clearly demonstrated that by adding as little as 0.125 μM of the FP probe, the enzymatic activity was completely stopped (Fig. 4). This experiment also demonstrated that only the serine type of hydrolase is presented in the formulated drug substance as the desthiobitin-FP probe is specific to serine hydrolase.

Depletion of Serine Hydrolase by Desthiobiotin- Fluorophosphonate (FP) Probe Results in a Diminished PS80 Degradation in Formulated mAb

We then performed depletion of serine hydrolase using immobilized ActivX FP serine hydrolase probe. The biotin part of the probe can be captured and immobilized on Streptavidin surface, allowing enrichment and purification of the captured serine hydrolases. The design of the depletion experiment serves two goals: 1) if PS80 degradation is caused by host cell proteins belonging to the serine hydrolase family, depletion will result in a diminished

Table 1
List of Identified HCPs Enriched from Native mAb-1 by FP Probe.

Protein in mAb-1 (Native)	Accession #	# Uniq.Peps.
Liver carboxylesterase B-1-like protein	A0A061I7X9	13
Liver carboxylesterase 1-like protein	A0A061IFE2	7
Peroxiredoxin-1	Q9JKY1	7
Cullin-9-like protein	A0A061IMU7	7
Junction plakoglobin	G3HLU9	6
Transthyretin	G3I4M9	6
Glyceraldehyde-3-phosphate dehydrogenase	A0A061IDP2	3
Ceruloplasmin	A0A061INJ7	3
Annexin	G3IG05	3
Anionic trypsin-2	G3HL18	3
Vitamin D-binding protein	G3IHJ6	3
Ubiquitin-40S ribosomal protein S27a-like protein	A0A061IQ58	3
Desmocollin-2-like protein	A0A061IEG0	2
tyrosine-protein kinase receptor	G3HG67	2
Actin, aortic smooth muscle	G3HQY2	2

Table 2
List of Identified HCPs Enriched from Denatured mAb-1 by FP Probe.

Protein in mAb-1 (Denatured)	Accession #	# Uniq.Peps.
Cullin-9-like protein	A0A061IMU7	7
Transthyretin	G3I4M9	4
Ubiquitin-40S ribosomal protein S27a-like protein	A0A061IQ58	3
Glyceraldehyde-3-phosphate dehydrogenase	A0A061IDP2	3
Peroxiredoxin-1	Q9JKY1	3
Vitamin D-binding protein	G3IHJ6	3
Ceruloplasmin	A0A061INJ7	2
tyrosine-protein kinase receptor	G3HG67	2
Desmocollin-2-like protein	A0A061IEG0	2
Liver carboxylesterase B-1-like protein	A0A061I7X9	2
Junction plakoglobin	G3HLU9	1

PS80 degradation; 2) the hydrolase captured on the Desthiobiotin-FP probe can be further identified by mass spectrometry analysis.

The depletion experiment was performed as outlined in the Material and Methods section with depletion scheme for mAbs shown in Fig. 5. Desthiobiotin-FP probe was coupled to Streptavidin Dynabeads for depletion of serine hydrolase. As shown in Fig. 5, prior to serine hydrolase depletion, approximately 44.7% of PS80 monoester degradation in mAb-1 was observed after 18 h incubation at 5 °C. Additional 18 h incubation led to complete PS80 monoesters loss. After serine hydrolase depletion, less than 8% PS80 degradation was observed after either 18 h or 36 h incubation. The depletion results demonstrated that the serine hydrolase(s) that degraded PS80 in mAb-1 was removed by the desthiobitin-FP probe. To ensure that it is the probe rather than streptavidin magnetic beads that interacted with the serine hydrolase, we performed the experiments by introducing a process control sample. The process control sample was produced by mixing mAb-1 with streptavidin magnetic beads only without adding desthiobiotin-FP probe. Approximately 29% and 86% of PS80 degradation in mAb-1 was observed after 18 h and 36 h 5 °C incubation, respectively, indicating that there were certain non-specific interactions between serine hydrolase and magnetic beads. Compared to the FP probe, the −serine hydrolase removed by the non-specific interactions was significantly less, therefore, the majority of the serine hydrolase was removed by specific binding between the immobilized FP probe and serine hydrolase.

Liver Carboxylesterases were Identified in the FP Probe-Enriched Fraction from mAb-1

The host cell proteins captured by the desthiobiotin-FP probe were subject to tryptic digestion and mass spectrometry analysis as described in the Materials and Methods section. Among the 15 host cell proteins identified (Table 1), CES-B1L and CES-1L were identified for the first time. By comparing with the denatured control sample (Table 2), we can conclude both proteins were captured by specific binding to the FP probe. CES-1L was only identified in the active form but not in the denatured form suggesting that it was biologically active in mAb-1. CES-B1L was identified in the active form with 13 unique peptides while only 2 unique peptides in the denatured forms, suggesting a small amount of CES-B1L protein was able to bind non-specifically to the magnetic beads, and the results were in line with process control results in the depletion experiment (Fig. 6). Nevertheless, the much higher number of unique peptides identified in the active forms suggests that CES-B1L is also responsible for PS80 degradation. For other identified host cell proteins, most can be easily determined as non-active enzymes as they were found in both conditions with a similar number of unique peptides, for example, cullin-9-like protein, glyceraldehyde-3-phosphate dehydrogenase and ceruloplasmin (Table 2). Anionic trypsin-2 was identified in the active form of mAb-1 as it is also a serine hydrolase, however, it can be excluded owing to its function as a protease irrelevant to PS80 degradation involving esterase bond cleavage.[20,21] The other two co-captured proteins, Junction plakoglobin and actin can also be excluded from degrading PS80 due to their lack of enzymatic functions. Interestingly, peroxiredoxin-1, transthyretin and annexin were identified in mAb-1 in the native state although they do not belong to serine protease family. Chen et al. discovered that 1-cys peroxiredoxin was a bifunctional enzyme with both peroxidase and phospholipase A2 activities.[22] As phospholipase A2 is also a serine hydrolase, therefore, it is possible that peroxiredoxin binds to the FP probe via its phospholipase active site. It was recently found that transthyretin possess metallopeptidase activity besides its well-known transport function via an Zn^{2+} induced active site composing of His-His-Glu triad similar to serine hydrolysis catalytic triad (Ser-His-Asp).[23,24] FP probe can also possibly enrich this protein by binding to its induced catalytic site and detect it. Annexin does not seem to contain any of these binding sites, however, it may bind to one of the phospholipases as it is a well-known phospholipase inhibitor.[25] To ensure these three HCPs do not possess or affect the PS80 degradation capability, we expressed the recombinant forms of these three HCPs and performed PS80 degradation experiment. As shown in Fig. 7, none of these three HCPs was able to degrade PS80. Taken together, the newly detected CES-B1L and CES-1L were most likely the cause for PS80 degradation. To determine the abundance of the newly identified hydrolases, CES-B1L and CES-1L, PRM analysis were performed. The concentrations of CES-B1L and CES-1L were determined to be 9.6 and 9.0 ppm relative to mAb-1, respectively. The relatively low

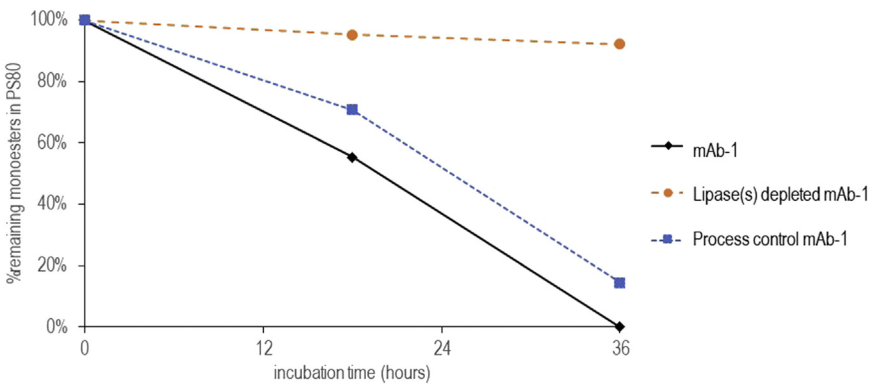

Fig. 6. Lipase depletion experiments, with original mAb-1, process control and lipase(s) depleted mAb-1 are indicated by filled diamond with black solid line, filled square with blue dotted line and filled circle with orange dashed line. The percentage of PS80 remaining is plotted against incubation time.

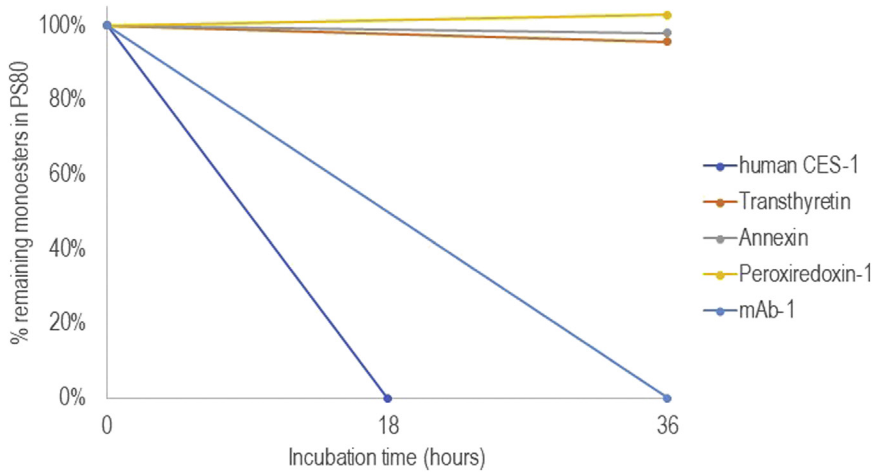

Fig. 7. Percentage of remaining PS80 monoesters incubated with 100 μg/mL of human CES-1, CHO transthyrtin, human annexin, CHO peroxiredoxin-1 and 50 mg/mL mAb-1 after 18 or 36 h at 5 °C.

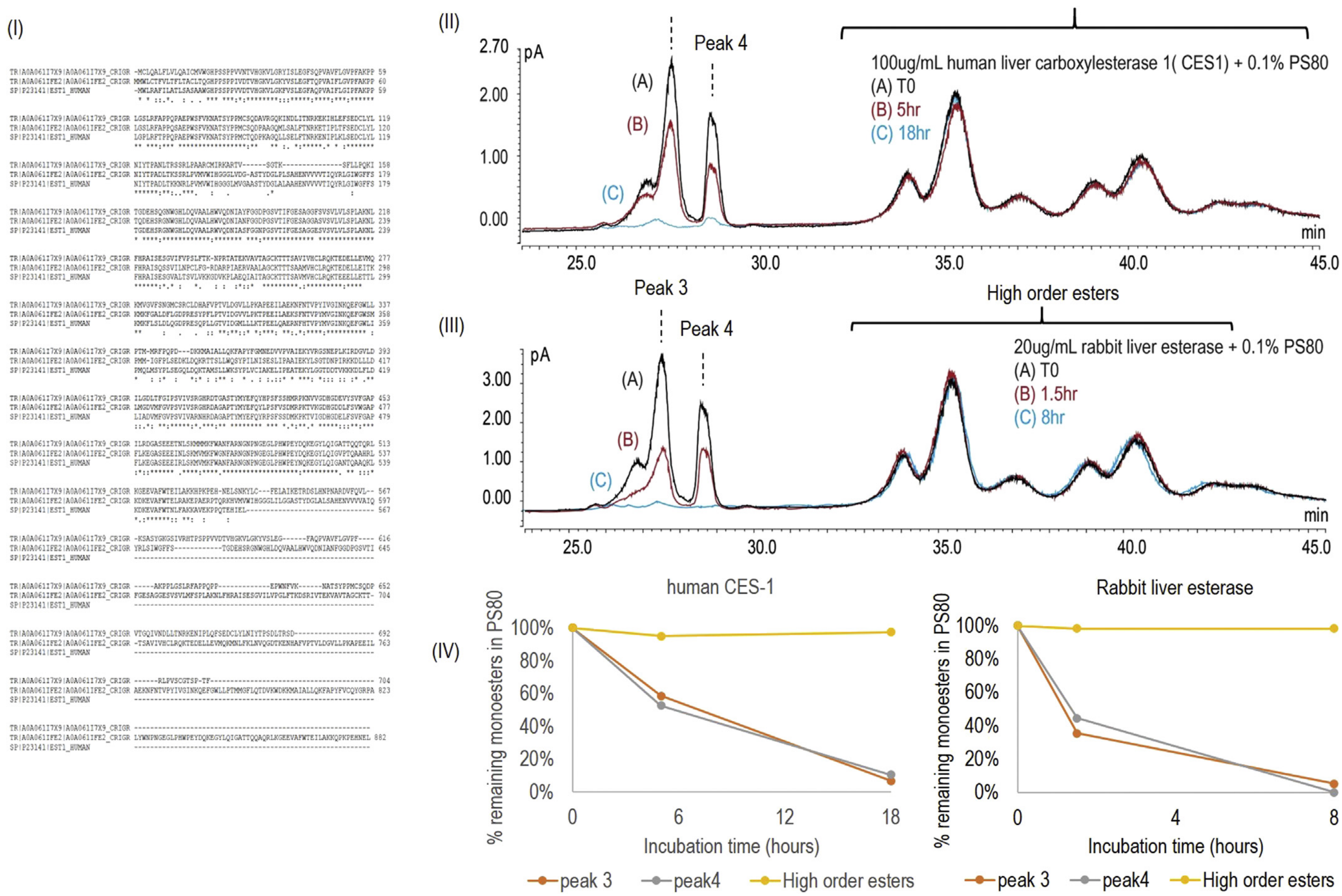

Fig. 8. (I). Sequence alignment of Liver Carboxylesterase B-1-like (A0A061I7X9), Liver Carboxylesterase 1-like (A0A061FE2) and Human liver carboxylesterase (hCES-1). (II). Chromatogram of 0.1% PS80 in 100 μg/mL commercial human liver carboxylesterases 1 incubated at 5 °C in 10 mM Histidine, pH6 for 0 h, 5 h and 18 h (III). Chromatogram of 0.1% PS80 in 20 μg/mL commercial rabbit liver esterase incubated at 5 °C in 10 mM Histidine, pH6 for 0 h, 1.5 h and 8 h. (IV). Left panel: Percentage of 0.1% remaining after 5 and 18 h of incubation with 100 μg/mL human CES-1 at 5 °C. Right panel: Percentage of 0.1% remaining after 5 and 18 h of incubation with 20 μg/mL rabbit liver esterase at 5 °C. Percentage of the initial area remaining for the representative peak 3 (POE sorbitan monooleate), peak 4 (POE isosorbide monooleate and POE monooleate) and sum of higher order esters (di-, tri-, tetra-esters) after hydrolysis is shown.

abundance of these two esterase suggests that their enzymatic activity for degrading PS80 is strong.

PS80 Degradation Pattern with Human Liver Carboxylesterase-1 and Rabbit Liver Esterase

One common and important practice in HCP analysis is to validate the function of enzymatic activities. The inhibition and depletion experiments have provided strong evidence that CES-B1L and CES-1L are most likely the hydrolase responsible for PS80 degradation. However, considering the FP probe we used is not specific to a single protein but to a family of proteins, it is possible that other hydrolase(s) presented in the drug substance that may also play a role in the degradation pathway. A spiked-in experiment can offer the essential verification on whether the suspected hydrolases are the root cause of PS80 degradation. If the spiked-in hydrolase can generate exactly the same degradation pattern as the endogenous hydrolase, the identified hydrolase can then be confirmed as the key element for PS degradation. This confirmation is usually difficult due to the lack of available active recombinant host cell proteins. To further confirm the role of these two newly identified hydrolases, we performed a BLAST search. The search results suggested commercially available rabbit liver esterase and human liver carboxylesterase 1 are functionally similar with each having 56.0% and 69.7% sequence homology to the first segment of CES-B1L and CES-1L, respectively (Fig. 8, I). Both human liver carboxylesterase 1 (Fig. 8-II) and rabbit liver esterase (Fig. 8-III) were chosen to compare the PS80 degradation pattern as mAb-1 (Fig. 3). Fig. 8-IV showed that both human and rabbit liver esterase exhibited an equivalent degradation pattern as that showed in mAb-1 (Fig. 3, right panel). In all three samples, the rapidly degraded components of PS80 were monoesters, which eluted between 25 and 30 min, while di-, tri- and tetra-esters which eluted after 32 min remained unchanged. The signature degradation pattern experiments by the known hydrolase verified that CES-B1L and CES-1L were responsible for PS80 degradation in mAb-1.

Conclusion

This study demonstrated the application of ABPP strategy to detect active hydrolase in mAb drug substances when they are present at such a low abundance level that immunoprecipitation based HCP profiling couldn't detect or is lack of coverage of them. This strategy not only can detect the low abundance host cell lipases, but also can effectively differentiate active serine hydrolase from the inactive ones. Through a combination of depleting the host cell proteins from mAb drug substances using the serine hydrolase-specific probe and measuring the polysorbate degradation of the serine hydrolase-depleted mAb, it is possible to determine whether the captured serine hydrolase is biologically active or not.

Our results showed that less than 10 ppm liver carboxylesterase presented in the drug substance was active enough to completely degrade monoesters in PS80 within one day at 5 °C. This is the first time these two unique host cell hydrolases have been reported to have hydrolytic activity on polysorbates in therapeutic protein product. Given HCPs as a critical quality attribute of the mAb product, and previous findings showed other lipase (Group XV lysosomal phospholipase A2 isomer X1 (LPLA2)) can degrade polysorbates at less than 1 ppm[7] it is very important to study these low abundance host cell proteins in the mAb product by developing orthogonal and targeted analytical methods. A comprehensive and deep readout of these critical HCPs will ultimately inform the process development group to design improved purification processes to eliminate them such that the long-term stability of the drug product can be guaranteed.

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